

The Faceted Plain-Text Lake Architecture

Abstract

This document specifies the architecture for a "Faceted Plain-Text Lake," establishing a foundational phase for modern knowledge management. It outlines the methodology for escaping the "Physical Shelf Limitation" of strict hierarchical taxonomies (like the Dewey Decimal System) by normalizing inbound data into Markdown and utilizing Polyhierarchical Faceting via YAML frontmatter. This architecture provides immediate searchability while laying the groundwork for future AI integration.

1. Introduction

Rigid hierarchical classification systems force knowledge artifacts into single-dimensional silos. A document covering "Automotive Manufacturing Processes" cannot easily reside in Engineering, Business, and Manufacturing categories simultaneously under traditional systems. The Faceted Plain-Text Lake decouples the document from the folder hierarchy, storing the artifact in a flat "lake" while preserving its multidisciplinary nature through embedded metadata.

2. Data Normalization Standard

To ensure universal machine readability and human accessibility, all ingested artifacts **MUST** be converted to a normalized text format. Markdown (RFC 7763) is the required format. The benefits include:

- **Portability:** Plain text is universally supported across storage systems and text editors.

- **LLM Compatibility:** Markdown provides syntactic structure (headers, lists, tables) that Large Language Models parse natively without requiring complex OCR or PDF extraction engines.

Any non-text media (images, audio, video) extracted during ingestion **MUST** be stored adjacent to the text artifact and linked via standard relative Markdown syntax.

3. Polyhierarchical Faceting via Frontmatter

To organize the unstructured lake, every document **MUST** include a YAML Frontmatter block at the beginning of the file. This block establishes *Faceted Classification*.

Instead of assigning a document a single path (e.g., /Engineering/Automotive/Manufacturing.md), the system assigns multiple contextual lenses. Below is the standard schema:

```
---
title: "Automotive Manufacturing Industry, Processes, and Manufacturing"
date_ingested: 2026-04-16
source_type: "Technical Manual"
document_id: "doc-8f7e6d5c"
facets:
  industry: "Automotive"
  domain: "Engineering"
  process: "Manufacturing"
  format: "Documentation"
tags: ["assembly line", "supply chain", "robotics"]
---
# Automotive Manufacturing Industry
[Document Content]
```

4. Retrieval and Traditional Search

Before implementing vector embeddings, this architecture supports robust retrieval using traditional inverted-index search engines (e.g., Elasticsearch, Apache Solr). Queries can be executed using Boolean logic against the defined facets (e.g., domain:Engineering AND process:Manufacturing), effectively simulating a dynamic, multi-dimensional taxonomy.